

chapter 1 (Introduction)

- Purpose of Database systems.
- view of Data □ Data Models.
- Data Definition language.
- Data Manipulation language.
- Transaction Management
- Storage Management
- Database Administrator.
- Database Users.
- Overall system structure.

First Chapter question (previous)

- ✓ ① What is database management system? Write down different application of DBMS? (SP-17)
- ✓ ② Define Data Model, compare Data Manipulation Language & Data Definition language. (SP-17)
- ✓ ③ Who is a Database administrator? What are the main functions of administrator? (Aut-19) / (SP-17)
- ✓ ④ Define DBMS, RDBMS & OODBMS. (Ans in 188) → Aut-1
- ⑤ Figure out the relation between DBMS, user and OS. How query processor processes query? Briefly explains the steps of the query processing. (Aut-19)
- ⑥ What are the necessities of Database management system? (SP-19)

✓ ⑦ List out different types of Database languages with their related benefits. (sp-19)

✓ ⑧ Define Different types of database model used till date. Which one is preferable. why? (Aut-19)

✓ ⑨ How could you hide information's of different levels from unauthorized users access? Explain with example (Aut-19)

✓ ⑩ Why modeling of database is necessary? point out the problem that could arise due to the lack of proper use of database model.

⑪ Define different functional components of a DBMS. explaining how a following query being executed through it.

Select first_name, salary
FROM employees. (sp-19)

⑫ compare among DDL, procedural DML, & Declarative DML. (sp-16)

✓ (13) What is data abstraction? Describe different levels of data abstraction. (SP-16)

✓ (14) Describe four advantages of DBMS over file system. (SP-14)

✓ (15) Describe database instance & schema with example? What are the functions of Storage Manager. (Aut-22)

✓ (16) Compare among Primary key, Super Key, candidate key & Foreign key with example. (Aut-22)

Answers Part

Q: What is database Management system? Write down different application of DBMS.

Ans: ☒ DBMS: A database Management System is a collection of interrelated data & a set of programs to access as accessible to those Data.

☒ Application of DBMS: $\rightarrow$ Databases are widely used. Here are some representative applications:—

♥ Enterprise Information:

Sales: For customer, product & purchase information.

Accounting: For payments, receipts, account balances, assets etc.

Human resources: For information about employees salaries, for generation of paychecks.

Manufacturing: For management of the supply chain & for tracking production of items in factories etc.

✶ **Banking & Finance:** Banking: For customer information; accounts, loans and banking transactions.

credit card transactions: For purchases on credit cards & generation of monthly statements.

Finance: For storing information about holdings, sales and purchase of financial instruments such as stocks & bonds.

✶ **Universities:** For student information, course registration & grades.

✶ **Airlines:** For reservations & schedule information. Airlines were among the first to use databases in a geographically distributed manner.

✶ **Telecommunication:** For keeping records of calls made, generating monthly bills & storing information about communication networks.

* * *

2. Q:- Define Data Model. Compare Data Manipulation Language & Data Definition Language.

Ans:→ **Data Model**: Data Model is the modeling of the data description, data semantics and consistency constraints of the data.

It provides the conceptual tools for describing the design of a database at each level of data abstraction.

Comparison Between DDL & DML:

DDL

1. A type of SQL command that helps to define database schemas.

2. Create, drop, alter are some DDL commands.

DML

1. A type of SQL command that helps to retrieve & manage data in relational databases.

2. Insert, update, delete & select are some DML commands.

2. Commands affect the entire database on the table

4. SQL statements cannot be rolled back

5. It helps us to define the structure of the Database.

6. They Don't have a further classification.

3. Commands affect one or more records in a table.

4. SQL statements can be rolled back.

5. It helps us to manage the data.

6. They are classified into procedural & non-procedural

* * *

Relational Model: The Relational Model is a data model in which data is organized into tables. A table is a collection of data records, each consisting of a collection of attributes. The data is organized into tables, each consisting of a collection of data records, each consisting of a collection of attributes.

Q. Define Different types of database model used till date. Which one is preferable? Why?

Ans: → ~~The~~ Different types of database model: → The data models can be classified into four different categories: →

1) Relational Model.

2) Entity Relationship Model.

3) Object Based Data Model.

4) Semistructured Data Model.

Relational Model: The relational model represents how data is stored in Relational Databases. A relational database consists of a collection of tables, each of which is assigned a unique name.

☐ Entity Relationship Model: The entity-relationship data model uses a collection of basic objects, called entities & relationships among these objects. An entity is a "thing"/"object" in the real world that is distinguishable from other objects.

☐ Object Based Data Model: An object data model is a data model based on object-oriented programming, associating methods with objects that can benefit from class hierarchies.

☐ Semi-structured data model: The semi-structured data model is a database model where there is no separation between the data & the schema. & the amount of structure used depends on the purpose.

* Preferable Data Model:— Relational

model is one of the most popularly used database model. In this model, changes in the database structure do not affect the data access. The revision of any information as tables consisting of rows and columns is much easier to understand. That's why Relational Data Model is more preferable.

* * *

4. Q: List out different types of Database languages with their related benefits.

Ans:— A database system provides a Data-Definition language to specify the database schema for a

Data Manipulation language to express database queries & updates. And another language is SQL Language.

• Data Manipulation language: A DML is a language that enables users to access data as organized by the appropriate data model.

The types of access are:-

- Retrieval of information stored in database.
- Insertion
- Deletion
- Modification

There are basically two types:

1) Procedural DMLs.

2) Declarative DMLs / nonprocedural DMLs

E.g. →

```
select instructor.name
from instructor
when instructor.dept_name = 'History';
```

- Data Definition Language (DDL): We specify a database schema by a set of definitions expressed by a special language called a DDL.

E.g. → create table account {

account - number

balance

integer(20),

☐ Benefits of DML: →

DML commands give us the flexibility to retrieve the data according to the requirements by applying the conditions using the WHERE clause.

☐ Benefits of DDL: - The single largest benefit of DDL is uniformity.

- SQL :- widely used non-procedural language.
E.g. :- find the name of the customer with customer id 192-83-7465

```
select customer.customer-name  
from customer.
```

```
where customer.customer id = 192-83-7465
```

□ Benefit of SQL :- SQL has the ability to function at a high operating speed.

Some other Data base languages are :-

→ DCL - Data Control language (grant, revoke)

→ TCL - Transaction control language.

* * *

Q: What are the necessities of Database management system?

Ans:- Necessities of DBMS:-

- 1) Improved efficiency
 - 2) Versatility
 - 3) Allowing categorization & structuring of available data.
 - 4) Multi access.
 - 5) Creating an organized working environment.
 - 6) Newer & better ways to manage data.
 - 7) Easy to store & access information
 - 8) Real-world entity
 - 9) Relational database.
 - 10) SQL
 - 11) Isolation of data & application.
 - 12) ACID properties.
 - 13) Object-oriented programming.
- After all, DBMS is general-purpose software that allow users to create, manipulate & design databases.

6. Q: Who is a database administrator? What are the main functions of a database administrator?

Ans: -

☐ Database administrator: A person who has such central control over the system is called a database administrator.

☐ Main functions of a administrator: →

1) Schema definition

2) Storage structure & access-method definition.

3) Schema & physical organization modification.

4) Granting of authorization for data access.

5) Routine maintenance.

* * *

7. Q: What is data abstraction? Describe different levels of data abstraction.

Ans: ☐ Data Abstraction: Data abstraction is the process of reducing a particular body

of data to simplified representation of the whole.

☐ Different level of data abstraction:—

Physical level: The lowest level of abstraction describes how the data are actually stored.

logical level: The next higher level of abstraction describes what data are stored in the database & what relationships exist among those data.

view level: The highest level of abstraction describes only part of the entire database. Even though the logical level uses simpler structures, complexity remains because of the variety of information stored in a large database / hide details of data types

8.9: Describe four advantage of Database over file system.

Ans. → Advantages over of Database over file system: →

1. **Data sharing**: The file system does not allow sharing of data. Whereas in DBMS, data can be shared easily due to a centralised system.
2. **Data Searching**: For every search operation performed on the file system a different application program has to be written. While DBMS provides inbuilt searching operation.
3. **Data integrity**: The file system does not provide any procedure to check constraints automatically. Whereas DBMS maintains data integrity.
4. **Data security**: File system provides a password to protect database but how long can the password in the case of DBMS this doesn't happen.

* * *

9. Why modeling of database is necessary?
point out the problem that could arise
due to lack of proper use of database
to model?

Ans: →

⊛ Modeling of Database is necessary.

Because, data models in DBMS help to
understand the design at the conceptual,
physical, and logical levels as it provides
a clear picture of the data making.

It is easier for developers to
create a physical database. Again

the data models are used to

describe how the data is stored,
accessed and updated in DBMS.

The problem that could arise due to lack of proper use of database model: →
poor database models leads to many problems down the line, such as sub-per part performance, the inability to make changes to accommodate new features and low quality data that can cost both time & money as the application ~~ex~~ evolves.

— 0 —

⑩ Describe database model instance & schema with example? what are the functions of storage manager?

Ans: →

Instance: — A collection of information that stored in a database at a particular moment is called an instance of database.

schema: The overall design of the database is called database schema. It will not be changed frequently. It is the logical structure of database.

Storage Manager: It is a program that provides an interface between the data stored in the database & the queries received. It also known as

Database control system.

- It maintains the consistency.
- It maintains integrity of the database by applying constraints.
- Executing the DCL statements.
- It is responsible for updating, storing, deleting & retrieving data in the database.

① Compare Among Primary key, super key, candidate key, & Foreign Key with examples.

Ans: primary key: The primary key of a relational table uniquely identifies each record in the table.

Ex:- The ideal primary key for a table of students would be their ID number, as this would uniquely identify each student in the table.

super key: A super key is a set of one or more keys that are used to identify data or records uniquely in a database table.

Ex: If each customer's email address is unique, the Email attribute can be a super key.

candidate key: A super key such that no proper subset is a super key within a relation.

EX: Emp_Id: An attribute that stores the value of the employee identification number.

Foreign key: An attribute or set of attributes within one relation that matches the candidate key of a relation.

EX: In a table of orders, a foreign key could be a reference to the primary key of the customers table, indicating which customer placed the order.

Tahiat